

Where Confidence Gets Laundered: Message versus Reader in a Model Handoff

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Abstract

How much of a sender’s uncertainty survives a handoff between frozen models, and where is it lost? We measure four handoffs from K2-Horizon-3.7B to 7B on 377 SQuAD items with clean / contradicted / answer-removed passages: the passage as text, the sender’s answer with a confidence word, number or candidate list, and its mapped KV cache. By AUROC of receiver first-token entropy against sender semantic entropy, the ordering is text (.597 / .615 / .671) $\gtrsim$ cache (.585 / .605 / .624) > confidence word (.552 / .542 / .592); cache—word excludes zero on one variant of three. Scored directly as a label the word reaches .955 / .942 / 1.000 (a tercile of that entropy), so the text arm’s loss is at this reader. Four-point probes do not localise the cache’s loss: the message probed at one position reads .557 / .579, the receiver .643 on clean passages. Within items (answer removed) receiver entropy tracks the sender’s through the cache (Spearman .280) and text (.322), barely through the word (.146), and not through another item’s cache (.004). The channel carries content: 63% on an 8-name binding test (chance 12.5%) with the wrong episode’s cache followed 64% of the time, and 53.0 SQuAD F1 against 54.7 for text. Score the message, then the reader: a one-line audit for any verbal-confidence baseline.

1. Introduction

When one language model hands work to another it usually writes a message: its answer, perhaps with a confidence word. Shi et al. (2026) call the result *confidence laundering*: the receiver treats a confident-looking guess as a fact. They compared text and latent handoffs by how well the receiver’s confidence tracks the sender’s. One motivation is an edge—

cloud stack that escalates from a small on-device model to a large one.

The measurement has a gap that, to our knowledge, no handoff study has closed. A low receiver-side score through a text handoff has two possible causes: the message does not carry the sender’s uncertainty, or the reader does not use it. Separating them costs one line of code—score the message itself against the same target that the receiver is scored against—and it changes what the comparison means. We call the first number the *message-level signal* and the second the *receiver-level signal*; their difference is the read-out gap. It is a difference of two AUROCs, not an information decomposition, and the cache has no label to score, so for the cache we use a linear probe and say so where we do.

This paper is that measurement, taken for four kinds of handoff between the same two frozen models (K2-Horizon-3.7B $\rightarrow$ 7B) on the same 377 SQuAD items with clean, contradicted and answer-removed passages: the passage as text; the sender’s answer plus a confidence word, a number, or its full candidate list; and the sender’s KV cache through a learned per-head map. Three controls back every number: content-free caches (another item’s, zeros, random) scored on each of the three passage variants; paired, passage-grouped bootstrap intervals on the three cache differences (cache minus word, minus question-only, minus text); and a within-item test where only the sender’s passage changes, so the question cannot be a common cause. Section 3 first shows that the cache carries the passage at all.

The findings are narrow. On clean and contradicted passages the receiver’s entropy tracks the sender’s best when it reads the passage, nearly as well through the mapped cache, and worst through a confidence word or number; on removed passages the candidate list matches the cache and question-only beats the word. The confidence word scored directly reaches .955 / .942 / 1.000, yet the receiver reading it lands at .552 / .542 / .592: for this base model, under one completion prompt with no instruction to hedge, the text arm’s loss is at the reader. The read-out failure is a property of the label channel, not of handoffs in general. For the cache arm, linear probes at four points of the chain do not localise the loss (Section 5); the parsimonious reading is that evidence crosses and the receiver recomputes, and no experiment

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here isolates the sender’s spread crossing as such. What the within-item test licenses is that the receiver’s confidence moves because of the cache’s content, nothing more about why.

Contributions. (1) A two-level report of uncertainty transfer—message-level and receiver-level signal against one sender target—for four handoffs, with content-free controls per variant, paired passage-grouped intervals on the cache differences, and a within-item intervention with a common-cause null (Sections 4 and 5). (2) The label-only audit, which relocates the text arm’s loss from channel capacity to read-out for this receiver (Section 5). (3) Linear probes at four points of the chain (sender state, mapped message, receiver state via cache and via text) against the sender-uncertainty target, which bound what a one-position read of the message carries and do not localise the cache arm’s loss (Section 5). (4) The instrument’s calibration: a six-arm binding test, a deranged-sender projector, layer-band injection, a continuation-retention measurement that confirms, for our own metric, the soft-prompt objection of Cheng et al. (2026), a cross-family pair and a failure pair, and the message’s cost in bytes and time (Sections 3 and 7). Shi et al. (2026) already compared answer-only, answer plus score, answer plus textual uncertainty and a latent carrier by AUROC against sender semantic entropy; we do not claim to be first to run that comparison. We have not found a handoff study that reports both levels (our literature pass ran to 2026-09-23); we report them with controls that survive a common-cause objection. Section 7 reports the message’s cost; at this size (72 MiB per 512-token prefix) the channel is a measurement instrument, not a deployment.

2. The Instrument

Models. Sender K2-Horizon-3.7B, receiver K2-Horizon-7B, both frozen, bf16, RoPE $\theta = 10^7$; they share a tokenizer and KV geometry (36 layers, 8 KV heads, head dimension 128). We capture the sender’s pre-RoPE K and raw V at every layer, map them, apply RoPE at the receiver, and insert the result as the receiver’s cache for the prefix positions. The receiver then reads only the question.

Projector. Stage 1: for each receiver layer ℓ and KV head h , a ridge regression from the concatenated $[K; V]$ of the sender (256-d per head) to the receiver’s $[K; V]$; source layers are the three sender layers whose K/V correlate best with receiver layer ℓ (“top-3”), which beat the aligned-layer map in a first pass. Stage 2: a gated SwiGLU residual (bottleneck 128, 151 M parameters) on top of the ridge output, trained with AdamW for 1,500 steps, effective batch 16 sequences of 1,024 FineWeb-Edu tokens, prefix 512, on the receiver’s cross-entropy over the continuation. Nothing in training mentions the tests.

Noma binding test. Each episode assigns one of eight single-token values to eight nonce names (“Noma is amber. Vela is indigo. ...”). Only the sender reads the facts. The receiver is asked “What colour is Vela?” and we score the eight values. Arms: *none* (question only), *text* (receiver reads the facts itself), *self* (receiver’s own cache re-injected; equals *text* to 10^{-4}), *raw* (sender cache injected without a map), *project* (mapped cache), *derange* (mapped cache from another episode with the same layout but a different value for the queried name). Content transfer = project – derange. For *derange* we also report the follow rate: how often the answer is the *other* episode’s value.

Passages. SQuAD v1.1 validation: the sender reads the passage (≤ 512 tokens), the receiver reads only the question and answers greedily (≤ 8 tokens, stopped at newline); token-F1 against the gold spans.

Uncertainty protocol: items and target. Following the knowledge-conflict and semantic-entropy protocols (Longpre et al., 2021; Xie et al., 2024; Farquhar et al., 2024), each SQuAD item gets three passages: *clean*, *contradicted* (one fluent sentence asserting a counter-answer, written by the 375B judge and kept only if DeBERTa-MNLI says it entails the counter-answer and not the gold: 587 of 941 survive), and *removed* (the gold sentence deleted). We keep the 377 items (344 distinct passages) on which the receiver reading the clean passage reaches token-F1 $\geq .5$. The sender’s uncertainty is its semantic entropy (SE) over 10 sampled answers (mean 0.98 / 1.18 / 1.96 over the 377 kept items on the three variants, so the sender does notice); the target is SE thresholded at its median per variant. Ten samples make SE coarse: 38 / 40 / 28 distinct values per variant, and on removed passages 22% of items sit exactly at the threshold. The receiver’s uncertainty is its entropy at the first answer token.

The four messages. Two are functions of the passage alone, $M = f(c)$, with the question unseen by the sender: *text* (the receiver reads the passage) and *mapped cache*. Two require the sender to answer the question, $M = f(c, q)$: *verbal* (the sender’s greedy answer and “confidence: high / medium / low”) and its variants with a number (the sender’s sample agreement, e.g. “0.8”, the intrinsic-confidence definition of Yona et al. (2024)) or the full candidate list (every answer cluster with its sampled frequency, e.g. “1929 (0.6); 1932 (0.3); 1935 (0.1)”). The four are therefore not one family of messages; the verbal arms have seen the question and the state arms have not, and we compare them anyway because that is the choice a system designer faces. The confidence word is an oracle label built from the sender’s *actual* SE terciles, favourable to the text channel; the sender does not verbalise its own confidence. The terciles are taken over all 587 items per variant and the target median over the 377 kept items, which is why the label’s own AUROC is

below 1 on two variants (Section 5). The receiver’s prompt is a plain completion—“A smaller model that read the passage answered: X (confidence: *high*).” followed by the question—with no chat template, no few-shot examples and no instruction to hedge; we did not ablate it, and Section 9 says what that leaves open.

Controls and metrics. Content-free controls: a *deranged cache* (another item’s mapped cache), a *zero cache*, a *moment-matched random cache*, and *question only*. Each handoff is scored at two levels against the same target u_S (sender SE above its median): the *message-level signal*, the AUROC of a fixed function of the message alone (a confidence word’s rank; for the cache, a linear probe on the mapped message), and the *receiver-level signal*, the AUROC of the receiver’s first-token entropy after reading it. Their difference is the read-out gap: a difference of two AUROCs, not an information decomposition, and the cache’s two levels use different instruments, so the gap is comparable within an arm, not across arms. The drop ratio is a handoff’s decrease in the receiver’s $P(\text{gold})$ from clean to removed divided by the text path’s decrease; it is a relative drop in confidence, not a fraction of uncertainty information retained, and the paths start from different confidence levels. Values are in Table 5.

Capacity, stated correctly. A message that is the argmax of the sender’s belief p_S plus a k -level label takes at most $|Y| k$ values, so it carries at most $\log |Y| + \log k$ nats about p_S ; a state message is not so bounded, although what the receiver recovers from it is bounded by the map and by its reading. This limits information about the *distribution* p_S (which alternatives, how spread), not a detector’s AUROC against a thresholded target: a three-level label reaches AUROC .955 against the median split (Section 5). An earlier draft of this paper deduced a lower receiver-level AUROC for the text arm from this “log 3 nats” bound; that deduction was wrong and is withdrawn. The bound is not what limits the text arm here, and no measurement in this paper tests it. *Why recovering the belief would matter.* If the receiver’s recovered belief $\tilde{b}$ is within total variation ε of the sender’s b and one-step rewards lie in $[0, R_{\max}]$, then acting on $\tilde{b}$ instead of b loses at most $2R_{\max}\varepsilon$ in expected reward. That is what a decision test should measure; this paper measures the read-out gap and the tracking of u_S , not regret.

3. Calibrating the Instrument: Content Must Cross

Before the receiver’s confidence can track the sender’s, the cache must carry the passage. This section establishes that with the controls that the audits of Cheng et al. (2026) and Zhang & Emu (2026) ask for, and shows why a continuation-retention headline cannot.

Binding transfer. On the 8-name test (Table 1) the trained projector reaches 63.0% (chance 12.5; the wrong episode’s cache 4.6); the easier 6-name variant ($n = 300$), used for the seeds, the ablations and Figure 1, gives 71.3 / 73.0 / 75.3%

Table 1. Binding transfer, 8 names $\times$ 8 values, $n = 1000$ episodes, chance 12.5%. All entries are percentages. “follow” is the rate at which the receiver, handed the deranged cache, answers with the other episode’s value; “content” is project – derange.

projector	none	text	raw	project	derange	follow	content
ridge top-3 (closed form)	10.6	100	14	21.7	11.4	18.2	10.3
+ residual, 1500 steps	10.6	100	14	63.0	4.6	63.7	58.4
residual, deranged sender	10.6	100	14	23.4	11.3	22.7	12.1

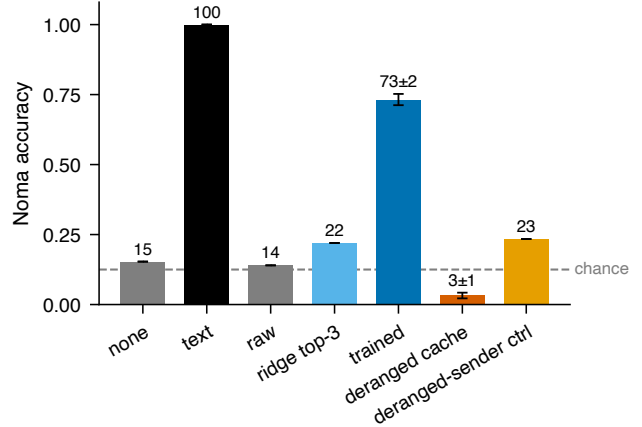


Figure 1. Noma accuracy by arm on the 6-name variant ($n = 300$). “trained” and “deranged cache” are the mean over three training seeds with error bars showing one standard deviation; the deranged-sender control is the *project* arm of the projector trained with a deranged sender ($n = 1000$, 8-name variant). Dashed line: chance (12.5%).

over three training seeds (mean 73.2, sd 2.0). Two ablations on the 6-name variant (Appendix B): freezing the ridge and training only the residual gives 59.0%, so letting the linear map move during training is worth ~ 14 points; replacing the correlation-selected top-3 source layers with the aligned-layer map gives 32.7%, so which sender layers feed each receiver layer is the largest design choice. Matching shape alone (*raw*) does nothing. Handed the wrong episode’s cache the receiver answers with the wrong episode’s value 63.7% of the time and its accuracy falls below chance: it is not ignoring the cache, it is reading it. The same protocol with cities and animals (8 names, $n = 500$) gives 61.8% and 67.2%, and padding the facts with up to 512 filler tokens does not reduce accuracy (Appendices C and E).

Deranged-sender projector. A projector trained with the identical recipe but with the sender reading text unrelated to the receiver’s continuation cannot learn a content channel, only a soft prompt. It reaches 23.4% on the 8-name binding test (Table 1, row 3). It is not content-free: its ridge base is weakly content-dependent (ridge alone 21.7%), so it controls for the *learned* part only, and we name it accordingly.

Where the binding lives. Table 2 (plotted in Appendix E): the first half of the network carries nothing usable on its own, the second half alone carries most of it, and no six-layer

Table 2. Where the binding lives. This episode’s mapped cache is injected only into a band of receiver layers, the deranged episode’s elsewhere. $n = 500$, 6-name variant; accuracy in percent.

layers	0–5	0–11	0–17	12–23	24–35	30–35	18–35	all
accuracy	7.2	8.0	10.4	28.6	30.8	7.6	51.8	63.0

Table 3. Real passages: SQuAD v1.1 validation, $n = 300$ (600 for the trained projector). Token-F1 against the gold spans; mean gold log-probability per token.

receiver sees	none	text	raw	ridge	trained	deranged-sender ctrl	trained, derange arm
F1	16.6	54.7	0.6	14.3	53.0 (51.8)	15.1	13.8
gold log-prob / token	−3.57	−0.90	−11.2	−3.88	−1.74	−3.77	−4.18

band suffices. The binding is read out from a distributed set of later layers; the filler used to complete the cache in these runs is the deranged episode’s cache, a plausible but wrong state.

Real passages. Table 3 and Figure 2 apply the same projectors, never trained on questions, to SQuAD. On F1 the trained projector recovers 96% of the gap between question-only and reading the passage (53.0 at $n = 300$; 51.8 at $n = 600$); on gold log-probability it recovers 69%. The wrong passage’s cache (13.8), the ridge map (14.3) and the deranged-sender projector (15.1) all sit at or below question-only (16.6). On passages, as on the binding test, the channel carries content, not a bias toward answering.

Continuation retention cannot tell a channel from a prompt. Following the closed-form-map literature (Heo et al., 2026) we also report retention (Table 4; plotted in Appendix H). Theirs is an accuracy ratio (transferred accuracy over the receiver’s own); ours is the fraction of the continuation-loss gap recovered on 128 fresh held-out sequences deduplicated against training text,¹ so the table critiques our own metric, not theirs. The deranged-sender projector—a learned soft prompt on a weakly content-dependent ridge base—recovers 77% of the oracle gap, and almost all of it (0.72) survives when its input is deranged at test time. The trained projector’s 1.23 decomposes into ~ 0.3 of the same prompt-like effect and ~ 0.9 that disappears when the sender’s text is swapped. Retention above 1 is not evidence of communication, as Cheng et al. (2026) objected; the binding test with its derange arm is. With that established, the instrument is calibrated: the cache carries the passage, and any uncertainty signal measured through it is measured through a channel that could in principle carry one.

¹On 256 such sequences: 1.19 for the trained projector, -0.38 for the ridge.

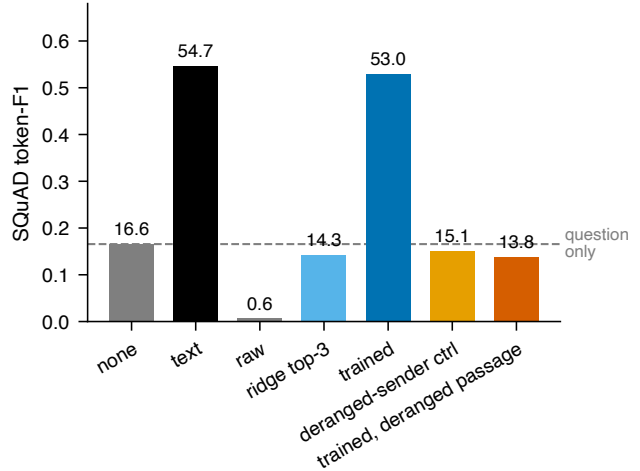


Figure 2. SQuAD token-F1 by arm ($n = 300$). Dashed line: question only. The trained projector sits within two points of the receiver reading the passage itself; the ridge map, the deranged-sender control, and the trained projector given the wrong passage’s cache all sit at or below question-only.

Table 4. Continuation retention is the wrong headline. Retention = $(\text{loss}_{\text{none}} - \text{loss}_{\text{project}}) / (\text{loss}_{\text{none}} - \text{loss}_{\text{oracle}})$ on 128 fresh held-out sequences; the second column swaps the sender’s text at test time.

projector	retention	retention, sender text deranged
ridge top-3	−0.30	−0.66
+ residual	1.23	0.31
residual, deranged sender	0.77	0.72

4. How Much Uncertainty Survives Each Handoff

Result. Table 5 and Figure 3 (left). On clean and contradicted passages the ordering of receiver-level signal is text, then the mapped cache, then every summary of the sender’s belief; on removed passages the candidate list matches the cache (.626 vs .624) and question-only (.603) beats the word (.592). The cache beats the confidence word by $+.03$ to $+.06$ AUROC, and its drop ratio is .54— $(.279 - .130) / (.437 - .162)$ —where the word’s is .30. Writing the confidence as a number does not help the text channel (.548 / .532 / .563; drop ratio .32), and neither does the strongest text message we can build from the sender’s samples, every answer cluster with its frequency (.495 / .489 / .626; drop ratio .44): the receiver’s confidence tracks the sender no better from it than from a word, except on removed passages. Intervals were computed for three differences, each a paired bootstrap over passages (344 passages, 2,000 resamples). Cache—word excludes zero on contradicted passages ($[-.008, .117]$) and not on clean ($[-.027, .094]$) or removed ($[-.024, .086]$); cache—question-only excludes

Table 5. Sender uncertainty through each handoff; 377 SQuAD items on which the receiver reading the clean passage reaches token-F1 $\geq .5$. AUROC is of the receiver’s first-token entropy against the sender’s semantic entropy (thresholded at its median per variant); the drop ratio is the receiver’s $P(\text{gold})$ loss from clean to removed relative to the text path’s. The two “scored directly” rows are the message-level signal: the label itself ranked against the same target, with no receiver (Section 5). Two projectors trained with uncertainty objectives are in Appendix A.

receiver sees	$P(\text{gold})$	entropy	AUROC vs sender SE	drop ratio
	clean / contra. / removed	clean / removed	clean / contra. / removed	
text	.437 / .290 / .162	1.14 / 3.03	.597 / .615 / .671	1 (def.)
text + answer + confidence word (verbal)	.253 / .193 / .171	2.88 / 2.80	.552 / .542 / .592	.30
text + answer + numeric confidence (sample agreement, e.g. “0.8”)	.258 / .197 / .170	2.93 / 2.93	.548 / .532 / .563	.32
text + all candidate answers with their sampled probabilities	.264 / .210 / .144	2.68 / 2.63	.495 / .489 / .626	.44
mapped cache	.279 / .249 / .130	1.78 / 3.16	.585 / .605 / .624	.54
deranged cache	.105 / .106 / .105	3.88 / 3.88	.536 / .488 / .564	0
zero cache	.034 / .033 / .037	6.41 / 6.18	.496 / .481 / .502	0
moment-matched random cache	.005 / .004 / .005	7.24 / 7.30	.457 / .472 / .580	0
question only	.127	3.69	.532 / .479 / .603	0
<i>message-level signal (no receiver)</i>				
confidence word, scored directly as a label	—	—	.955 / .942 / 1.000	—
numeric confidence, scored directly	—	—	.870 / .848 / .735	—

zero on contradicted ($[-.065, .180]$) and is borderline on clean ($[-.005, .111]$); cache—text includes zero on clean ($[-.066, .047]$) and contradicted ($[-.064, .047]$) and excludes it on removed ($[-.091, -.002]$), where the text path is above the cache. The direction is consistent; the size is modest and established on one variant of three.

Controls, per variant. The content-free controls do not sit at one level. On clean and contradicted passages they lie at .46–.54 (question only .532 / .479; deranged .536 / .488; random .457 / .472; zero .496 / .481) against the cache’s .585 / .605. On removed passages question-only alone gives .603, the random cache .580 and the deranged cache .564 against the cache’s .624: the question already carries most of the signal when the answer is gone, and question-only there beats the word (.592). The cache’s increment over question-only is therefore +.053 on clean, +.126 on contradicted and +.021 on removed passages, and only the middle one is established.

Behaviour under contradiction and removal. When the answer is removed, the cache path loses 54% of the confidence the text path loses and its entropy rises by 73% of the text path’s rise. Under a contradiction the text path is swayed slightly more by the inserted sentence than by the original ($P(\text{counter}) .32 > P(\text{gold}) .29$), while the cache path keeps the original ahead ($.25 > .22$); across all conditions the cache path is less confident than the text path on clean passages ($P(\text{gold}) .28$ vs .44), so a receiver that treats the two channels as equivalent would under-trust the cache (Figure 9). What this section cannot say is why the ordering is what it is; that is the next section’s job.

5. Where the Loss Occurs

(a) The label carries the signal; this receiver does not read it. The confidence word is built from the sender’s own

SE terciles, so scored as a label it predicts the binary target almost perfectly without any receiver: AUROC .955 / .942 / 1.000 on the three variants, the numeric label .870 / .848 / .735 (Table 5, bottom). These numbers are a property of the construction—a tercile split over all 587 items against a median split over the 377 kept items—and the 1.000 on removed passages is a degeneracy: 22% of items tie at the threshold and the tercile boundary coincides with it. The audit is a sanity check, not a finding; the finding is the other bar: the receiver’s first-token entropy after reading that word reaches .552 / .542 / .592. The text channel is not capacity-limited here; this base-model receiver, under one completion prompt, does not turn a label into its own spread. That is a statement about one reader and one prompt format; a receiver instructed to hedge, given few-shot examples, or instruction-tuned could erase the contrast, and we did not test one. It is also why the capacity argument of our earlier draft (box, Section 2) cannot be the mechanism of the verbal arm’s shortfall.

(b) Within-item intervention. Cross-item correlations admit a common cause: a hard question makes both models uncertain whether or not anything crosses. The variants allow a paired test with the question held fixed and only the sender’s passage changed (clean $\rightarrow$ removed; the sender’s SE rises by 0.98 on average). Per item, the Spearman correlation between the receiver’s entropy change and the sender’s SE change is .280 through the mapped cache, .322 when the receiver reads the text, .146 through the confidence word, and .004 through a cache from another item, which shares the question and rules out the common cause (Figure 3, right). The same-sign rates (88 / 90 / 49 / 47%) are weaker evidence than they look: the sender’s SE rises on 91.5% of items, so any arm whose entropy simply rises on removal scores near that ceiling and an arm with no mean shift sits at a coin flip; same-sign measures the mean shift, the Spearman the

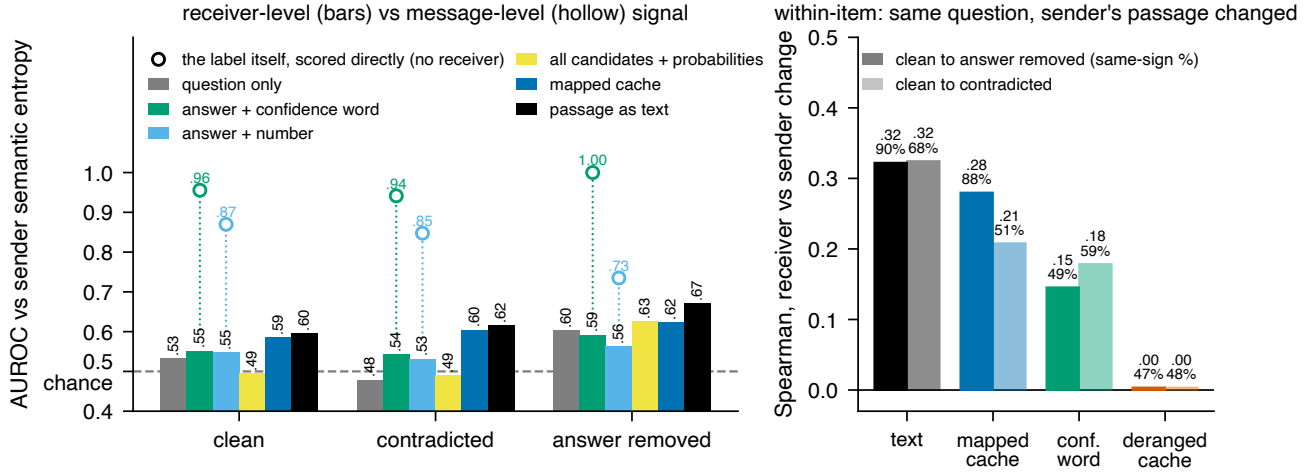


Figure 3. Left: receiver-level signal (bars: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, 377 items) for six handoffs per passage variant, with the message-level signal of the label itself drawn as a hollow marker above the word and number arms; the vertical distance from marker to bar is the read-out gap. The cache has no label and no marker. Paired, passage-grouped intervals on the three cache differences are in the text. Right: within-item intervention, the question held fixed and the sender’s passage changed; bars are the Spearman correlation between the receiver’s entropy change and the sender’s semantic-entropy change, annotated with the same-sign rate, which is bounded by the 91.5% of items on which the sender’s entropy rises (Section 5).

per-item covariation. Under a contradiction the effect is smaller (cache .208, text .325, word .179, deranged .004). The receiver’s confidence moves with what the sender saw, and it moves because of the cache’s content. It licenses no more than that: the text path tracks at least as well, so the parsimonious reading is that the evidence crosses and the receiver recomputes its own uncertainty from it.

(c) Shape of the belief. Sender SE and receiver first-token entropy are different measurements; a diagnostic closer to the belief itself is the *shape*: per item with at least two sampled answer clusters, the Spearman correlation between the sender’s cluster frequencies and the receiver’s first-token mass on those candidates. On clean passages ($n = 323$) it is .199 through the mapped cache, .193 from the text, .166 from the explicit candidate list, and .052 / .042 for the deranged cache and question only: there the receiver recovers the sender’s alternatives about as well from the cache as from the passage, and no better from a list of them. Under a contradiction ($n = 351$) the order reverses (cache .153, list .201, text .234), and on removed passages every arm is near zero. No interval was computed for these correlations, and they are small enough that the diagnostic separates arms from the controls, not from each other.

(d) Four-point probes. The same standardised logistic probe ($C = 0.05$, five passage-grouped folds, no inner selection; the semantic-entropy-probe protocol of Kossen et al., 2024) for the binary sender-SE target at four points of the chain, on inputs of 2560 / 2048 / 6144 / 4096 dimensions and on the 587 items (the full set before the F1 filter, so these AUROCs are not directly comparable with Table 5). Clean / removed: the sender’s own last-prompt-token state,

averaged over layers 12 / 18 / 24, gives .739 / .706; the mapped message at its last prefix position, averaged over layers and heads, .557 / .579 (with the question-only receiver state concatenated, .563 / .603); the receiver’s state after reading through the mapped cache .643 / .563; after reading the text .650 / .654; the question alone .530 / .591. Three things follow. The sender holds more about its own uncertainty than any receiver state recovers. On clean passages the receiver reaches .643 through the cache, above the .557 that the probed message carries, so the message carries more than a one-position probe reads, and the cache leaves the receiver with as much as the text does (.643 vs .650); on removed passages the receiver (.563) falls below the probed message with the question (.603) and below the text (.654). These probes therefore do not localise the loss. Two caveats bound them further. The message-level probe sees one position of 512, so .557 / .579 bounds what one position carries, not what the map preserves; a pooled read-out of the whole prefix is the obvious next probe. And the sender probe saw the question as well as the passage, while the mapped message is passage-only, so sender-minus-message mixes question conditioning with map loss.

6. Two Objectives That Did Not Move the Loss

If the cache arm’s shortfall were an artefact of the projector’s training target, changing the target should move it. Two receiver-side objectives did not (Appendix A): a KL term to the receiver’s own text-path distribution leaves the drop ratio at .56 (vs .54) and AUROC within .02 (.590 / .589 / .611 vs .585 / .605 / .624) while costing 2 F1, and matching the receiver’s per-token entropy through the cache to the sender’s

Table 6. What the message costs. Bytes per prefix token of the mapped cache (512-token SQuAD passages; “KB” = 1024 bytes). Noma: 6-name variant, $n = 300$, seed-2 projector. “filler” fills the untransmitted layers or heads with the receiver’s own cache of a neutral text of the same length. The full sweep, including zero and mean-vector fills and smaller projectors, is in Appendix F.

message	KB / token	Noma	SQuAD F1
full mapped cache, bf16	144	75.3	53.0
int4 fake-quantised	36	75.0	51.6
layers 18–35 only, filler elsewhere	72	28.7	16.1
4 of 8 heads, filler elsewhere	72	26.7	12.8

own leaves it at .51 and costs 8 F1. Neither target contains the sender’s spread; a sender-side reader $r(m, q)$ trained on the sender’s candidate distribution and kept separate from the receiver’s own decision is the untested fix.

7. What the Channel Costs and Where It Works

Cost. The full mapped cache is $36 \text{ layers} \times 8 \text{ heads} \times 2 \times 128 \times \text{bf16} = 144 \text{ KB}$ per token, 72 MiB for a 512-token prefix, against $\sim 1 \text{ KB}$ of text; the projector has 151 M parameters. Table 6 (6-name binding variant, $n = 300$): int4 fake-quantisation keeps binding accuracy (75.0 vs 75.3%) and SQuAD F1 (51.6 vs 53.0) at a quarter of the bytes, but this is fake quantisation, not a packed transport. Nothing else is free. Dropping layers or heads collapses transfer even when the untransmitted layers hold the receiver’s own cache of a neutral text, a valid episode-free state: layers 18–35 alone give 28.7% on the 6-name variant and 16.1 F1, 4 of 8 heads 26.7% and 12.8 F1. Zero and mean-vector fills collapse the same way (Table 13), so the loss is missing content, not broken attention, and with this projector the message cannot be cut by layers. Low-rank truncation keeps the 6-name binding test (rank 32 / 16: 75.0%) but not passages (F1 45.3 / 29.6; Table 13). Latency on one A100: the receiver re-reading 512 tokens takes 48 ms, the projector 24 ms. But at 72 MiB per message a 1 Gbps link needs $\sim 604 \text{ ms}$ (int4: $\sim 151 \text{ ms}$) against 24 ms of receiver compute saved, so the compute advantage does not carry over to a network handoff at this size. A reader who can get the sender’s uncertainty by re-reading 1 KB of text should; the channel is worth its bytes only where the sender’s state, not its passage, is the point, and this paper has not exhibited such a case.

Where it works. The recipe crosses one family boundary and fails on one same-family pair (Appendix C). With Qwen3-4B as the sender (matched KV geometry, different vocabulary and pre-training; receiver positions aligned by character offsets) the trained projector reaches 76.7 / 70.3 / 68.0% on the 6-name binding test over three training seeds (mean 71.7, sd 4.5; deranged $\leq 4.7\%$, follow rate 80%

for seed 0) and SQuAD F1 48.0 / 51.9 / 44.9 (mean 48.3) against 54.7 for text, while the ridge map alone gives 18.0%: cross-family transfer is as strong as within-family on the binding test and within 7 F1 on passages. With the roles reversed (7B sender, 3.7B receiver) the pattern repeats at 63.3% on the 6-name variant. For K2-Horizon-0.9B, which differs from the receiver in vocabulary, layer count and head dimension (64) at once, the trained projector reaches 17.0% on the 6-name variant (deranged 14.0%, follow 11.3%) and 24.2 F1; this parameterisation fails on that pair, and which difference matters is untested. On HotpotQA bridge questions, where sender and receiver hold different paragraphs (Appendix D), the receiver’s F1 goes from 30.1 with its own paragraph to 39.9 ± 2.1 with the sender’s paragraph as a mapped cache, against 45.8 as text and 29.9–32.1 for a deranged cache. A handoff-policy experiment on SQuAD (Appendix G) is negative for a reason specific to the task and is summarised in Section 9.

8. Related Work

Confidence Laundering (Shi et al., 2026) is the position this paper measures: it compared answer-only, answer plus score, answer plus textual uncertainty and a latent carrier by AUROC against sender semantic entropy and found the latent carrier ahead; its one experiment (Qwen-1.5B in a HotpotQA tool loop, 20-sample semantic-entropy labels, probes at blocks 7 / 14 / 28) scores the downstream component only and reports no deranged or content-free control. Two causal audits supply the controls we adopt: Cheng et al. (2026) find that on natural benchmarks most of a relayed cache’s gain is receiver-side computation (MedQA: +0.4 of +14.7 points is content-specific, 90% CI -1.34 to $+2.14$), while content transfer appears only when the receiver needs sender-private information (+61 to +77 points); our derange and deranged-sender arms are built for that objection. Zhang & Emu (2026) decompose a latent channel’s effect into “sender content” and “any message present” with boundary-replacement controls, of which our deranged-sender projector isolates the second term. The channel literature around us has no message-versus-reader split: C2C (Fu et al., 2025) fuses sharer caches of 0.5B–4B, including a Llama→Qwen pair, into a 0.6B receiver through a trained fuser; Latent Cache Flow (Rossi et al., 2026) trains a small adapter (13–107 MB) on a Qwen3-0.6B pair and introduces the partitioned-HotpotQA setting we reuse in Appendix D; a closed-form KV map (Heo et al., 2026) reports 44–98% accuracy retention (HellaSwag) across same-family pairs; K-V cache alignment (Dery et al., 2026) trains adapters that bring Gemma-2 models’ caches into a shared space; a universal context-reuse layer (Li et al., 2026) makes one model’s KV state consumable across scale, tokenizer and family, scored on accuracy, perplexity and latency; StateBridge (Peng et al., 2026) aligns final-layer hidden states into the receiver’s in-

put space with a closed-form orthogonal map; Interlat (Du et al., 2025) has agents communicate entirely in latent space and argues latents keep several hypotheses; Dense Latent Communication (Chen et al., 2026) sends latent messages across heterogeneous agents and separates context-aware from context-unaware receivers; Dual-Cache (Liu et al., 2026a), CacheBridge (Qu et al., 2026), HyLaT (Mou et al., 2026) and LatentPort (Villani, 2026) extend cache transfer to heterogeneous pairs, hybrid protocols and recurrent state; CIPHER (Pham et al., 2023) had agents debate through embeddings. MentorPulse (Liu et al., 2026b) finds that a static one-shot latent handoff can fall below no guidance on long-form generation, a scope limit our short-answer setting does not probe; the alignment ladder of Marincat (2026) runs a leakage-controlled audit close in spirit to our derange arms; the Replay Gap (Gonuguntla, 2026) asks for branched rollouts, which remain a next step. On the uncertainty side we borrow the knowledge-conflict construction of Longpre et al. (2021) and Xie et al. (2024), semantic entropy as the sender’s uncertainty (Farquhar et al., 2024) and the hidden-state probe for it (Kossen et al., 2024); verbalised confidence can be better calibrated than token probabilities for RLHF-tuned models (Tian et al., 2023) but does not faithfully express a model’s sample-based uncertainty (Yona et al., 2024); distillation-style maps have been observed to contract predictive variance (Cui & Pei, 2026), which we cite as an observation to compare against, not as a mechanism for our map. The cache-transfer and audit papers from 2026-05 onward (Rossi et al., 2026; Zhang & Emu, 2026; Heo et al., 2026; Cheng et al., 2026; Li et al., 2026; Peng et al., 2026; Shi et al., 2026) are concurrent with ours; C2C, Interlat, K-V cache alignment, CIPHER and the uncertainty literature are prior work we build on.

9. Limitations

The uncertainty measurement rests on 377 items of one dataset, one sender–receiver pair, one projector seed and one sender-uncertainty measure; the other two projector seeds have not been run through the confidence protocol. Our pre-registered criterion (design note of 2026-09-22, in the repository) asked for cache AUROC $\geq .70$ and drop ratio $\geq .5$; the second was met (.54), the first was not (.585 / .605 / .624), so the paper reports the pre-registered fallback—content transfers, uncertainty only partly. The word moves the receiver’s AUROC by only $+.020 / +.063 / -.011$ over no message at all. The contested effect is small: cache minus word is .03–.06 AUROC (intervals in Section 4), and on removed passages question-only beats the word and text beats the cache. The absolute AUROCs are ~ 0.6 for every channel including text, so every claim is relative. “The receiver does not read the label” is a property of one base model under one completion prompt: no prompt ablation, no few-shot examples, no chat template, no instruction-tuned

receiver, and the candidate-list arm scoring below question-only on clean passages (.495 vs .532) says the text baselines are not the strongest a competent engineer could build. The four-point probes read one position of the mapped message, were run on the 587-item set rather than the 377, and do not localise the loss. The belief-shape correlations (.199 / .193 / .166) separate arms from controls, not from each other. Three controls are weaker than their names: the deranged-sender projector is not content-free (23.4% against a ridge base of 21.7% on the 8-name test), the retention critique is of our own metric rather than of Heo et al. (2026), and the layer-band experiment uses the deranged episode’s cache as filler. The setting is context reuse—the sender’s cache of a passage prefix—not a mid-task handoff of an agent’s reasoning or tool state, and nothing here says whether any of it survives a sender that has reasoned or generated at length (Liu et al., 2026b). The channel is expensive in bytes (144 KB per token; 604 ms per message on a 1 Gbps link) and the handoff-policy experiment (Appendix G) is negative on SQuAD because the receiver reading the passage is only 3.5 F1 above the sender (55.7 vs 52.2), so no trigger, not even a .749-AUROC probe on the sender’s hidden state, makes a handoff pay; the deployment that motivates the paper does not exist at this message size.

10. Conclusion

Score the message, then score the reader. Done for four handoffs between two frozen models, that pair of numbers turns “a text handoff launders the sender’s uncertainty” into something narrower and testable: for this receiver, a confidence word carries the signal (.955 / .942 / 1.000) and the reader discards it (.552 / .542 / .592), while a mapped KV cache with no label at all leaves the receiver tracking the sender nearly as well as reading the passage on clean and contradicted passages (.585 / .605 vs .597 / .615; intervals include zero) and significantly below it on removed ones (.624 vs .671; $[-.091, -.002]$), and moves its confidence within an item because of the cache’s content (Spearman .280 vs .004 for another item’s cache). Where the cache arm loses what it loses, our probes do not say: a one-position probe of the message reads .557 / .579, below what the receiver recovers on clean passages (.643). The measured ordering is text $\gtrsim$ cache $>$ label $>$ deranged; the cache-minus-label gap is small and established on one variant of three; and the honest reading of the cache arm is that evidence crosses and the receiver recomputes. What would overturn the paper’s main claim—that this reader discards a label it was handed—is a receiver whose entropy follows “confidence: low”: an instruction-tuned or few-shot reader. That is the first experiment we would run. What would make the channel worth its bytes is a case where the sender’s state, not its passage, is the point; we have not exhibited one.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here. One consideration specific to cache handoff: a cache that a receiver “believes” (Section 3) is also a cache that can be tampered with; deployments that move caches between devices should authenticate them as they would any other input.

References

- Chen, S., Zhang, X., Wu, M., Tremblay, J., Blukis, V., Birchfield, S., Vidal, R., Velasquez, A., Liu, S., and Qu, Q. See what i see, know what i think: Dense latent communication across heterogeneous agents. *arXiv preprint arXiv:2606.13594*, 2026.
- Cheng, J., Das, S., and Ramnath, R. When does latent communication pay? a causal audit of relayed KV caches in multi-agent LLMs. *arXiv preprint arXiv:2608.04893*, 2026.
- Cui, Z. and Pei, J. How is uncertainty propagated in knowledge distillation? *arXiv preprint arXiv:2601.18909*, 2026.
- Dery, L. M., Yahav, Z., Prior, H., Feng, Q., Shen, J., and Szlam, A. Latent space communication via K-V cache alignment. *arXiv preprint arXiv:2601.06123*, 2026.
- Du, Z., Wang, R., Bai, H., Cao, Z., Zhu, X., Cheng, Y., Zheng, B., Chen, W., and Ying, H. Enabling agents to communicate entirely in latent space. *arXiv preprint arXiv:2511.09149*, 2025. ACL 2026.
- Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.
- Fu, T., Min, Z., Zhang, H., Yan, J., Dai, G., Ouyang, W., and Wang, Y. Cache-to-cache: Direct semantic communication between large language models. *arXiv preprint arXiv:2510.03215*, 2025. ICLR 2026.
- Gonuguntla, A. The replay gap: Static evaluation of model switching in LLM agents scores the wrong world. *arXiv preprint arXiv:2608.08239*, 2026.
- Heo, T., Shafipour, R., Zhao, R., Golub, M., Kamani, M. M., Borkar, R., Chandran, M. T., Zardoshti, P., and Rouhani, B. D. Cross-model KV cache transfer in LLM families: A closed-form linear mapping for prefill reuse. *arXiv preprint arXiv:2608.03893*, 2026.
- Kossen, J., Han, J., Razzak, M., Schut, L., Malik, S., and Gal, Y. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint arXiv:2406.15927*, 2024.
- Li, Y., Jiang, D., Zhao, Y., and Li, B. A universal context-reuse layer for cross-model KV sharing. *arXiv preprint arXiv:2608.30963*, 2026.
- Liu, J., Zhang, Q., Jia, Y., Kan, Z., and Wang, S. Dual-cache latent space communication between heterogeneous language models. *arXiv preprint arXiv:2608.20617*, 2026a.
- Liu, Z., Li, G., Qiu, C., Kong, W., and Kalnis, P. MentorPulse: Refreshing cross-model latent guidance for long-form generation. *arXiv preprint arXiv:2608.20927*, 2026b.
- Longpre, S., Perisetla, K., Chen, A., Ramesh, N., DuBois, C., and Singh, S. Entity-based knowledge conflicts in question answering. In *Proceedings of EMNLP*, 2021. arXiv:2109.05052.
- Marincat, N. Portable semantics, private dialects: Reuse and negative transfer in latent communication between language-model cells. *arXiv preprint arXiv:2609.11365*, 2026.
- Mou, X., Wang, S., Li, Z., He, Y., and Wei, Z. HyLaT: Efficient multi-agent communication via hybrid latent-text protocol. *arXiv preprint arXiv:2605.25421*, 2026.
- Peng, Y., Zhang, D. C., Wang, X., and Aletras, N. State-Bridge: Training-free hidden-state alignment for latent communication in LLM multi-agent systems. *arXiv preprint arXiv:2608.13317*, 2026. COLM 2026.
- Pham, C., Liu, B., Yang, Y., Chen, Z., Liu, T., Yuan, J., Plummer, B. A., Wang, Z., and Yang, H. Let models speak ciphers: Multiagent debate through embeddings. *arXiv preprint arXiv:2310.06272*, 2023.
- Qu, X., Lu, S., Chen, Z., Wang, S., and Lin, T. CacheBridge: Efficient cross-model KV cache transfer. *arXiv preprint arXiv:2609.00891*, 2026.
- Rossi, M., Raghunath, P., and Wu, E. Latent cache flow: Model-to-model communication without text. *arXiv preprint arXiv:2605.22863*, 2026.
- Shi, K., Zhang, Z., Bao, H., Nelson, C., and Ye, Y. Confidence laundering in agent systems: Why uncertainty needs a latent carrier. *arXiv preprint arXiv:2606.20662*, 2026.
- Tian, K., Mitchell, E., Zhou, A., Sharma, A., Rafailov, R., Yao, H., Finn, C., and Manning, C. D. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of EMNLP*, 2023. arXiv:2305.14975.

- Villani, S. P. LatentPort: Beyond KV cache – cross-model transfer of recurrent memory in hybrid language models. *arXiv preprint arXiv:2609.25053*, 2026.
- Xie, J., Zhang, K., Chen, J., Lou, R., and Su, Y. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In *Proceedings of ICLR*, 2024. arXiv:2305.13300.
- Yona, G., Aharoni, R., and Geva, M. Can large language models faithfully express their intrinsic uncertainty in words? In *Proceedings of EMNLP*, 2024. arXiv:2405.16908.
- Zhang, H. and Emu, M. Do latent channels actually communicate? a causal audit of latent multi-agent LLM. *arXiv preprint arXiv:2607.26773*, 2026.

A. Training Objectives That Did Not Recover the Sender’s Spread

Table 7. Two projectors trained with an uncertainty objective (weight 1, 1,000 steps), scored exactly as in Table 5.

receiver sees	$P(\text{gold})$	entropy	AUROC vs sender SE	drop ratio
	clean / contra. / removed	clean / removed	clean / contra. / removed	
mapped cache (reference, Table 5)	.279 / .249 / .130	1.78 / 3.16	.585 / .605 / .624	.54
mapped cache, KL-trained projector	.272 / .241 / .118	2.38 / 3.76	.590 / .589 / .611	.56
mapped cache, sender-entropy-matched projector	.247 / .222 / .107	2.23 / 3.64	.596 / .587 / .627	.51

Adding $\text{KL}(\text{text-path} \parallel \text{cache-path})$ on the receiver’s next-token distributions to the projector’s objective (weight 1, 1,000 steps) raises the cache path’s entropy everywhere (clean 2.38 vs 1.78) without making it track the sender better (AUROC .590 / .589 / .611; drop ratio .56 vs .54) and costs 2 F1. Matching the receiver’s own text-path distribution is the wrong target: what is lost is the *sender’s* spread, which the receiver’s text path does not contain either. A sender-side target—matching the receiver’s per-token entropy through the cache to the sender’s own per-token entropy on the training text—was the second attempt (weight 1, 1,000 steps): binding accuracy holds (72.3%; Appendix B), SQuAD F1 falls to 45.3, and the uncertainty measures do not move (AUROC .596 / .587 / .627; drop ratio .51). Entropy matching is not belief matching; the untested fix is a separate reader $r(m, q)$ trained on the sender’s candidate distribution.

B. Ablations on the 6-Name Variant

All runs: sender K2-Horizon-3.7B, receiver K2-Horizon-7B, colour attribute, 6 names, $n = 300$ episodes, chance 12.5%. Accuracy in percent. The two 1,000-step rows with an added uncertainty term (KL, sender-entropy matching) use the full 151 M-parameter projector.

Table 8. Projector ablations on the 6-name Noma variant.

projector	project	derange	follow
ridge top-3 (closed form)	22.0	10.7	30.3
+ residual, 300 steps	56.3	6.3	53.7
+ residual, 1500 steps, seed 0	71.3	4.3	71.7
+ residual, 1500 steps, seed 1	73.0	3.0	74.0
+ residual, 1500 steps, seed 2	75.3	2.3	76.0
+ residual, 1500 steps, ridge frozen	59.0	7.0	59.3
+ residual, 1500 steps, aligned-layer ridge	32.7	8.0	34.7
+ residual, 1000 steps, bottleneck 32	64.3	5.7	62.3
+ residual, 1000 steps, bottleneck 16	61.3	7.7	60.0
+ residual, 1000 steps, + $\text{KL}(\text{text} \parallel \text{cache})$	66.0	5.3	63.3
+ residual, 1000 steps, + sender-entropy matching	72.3	3.0	73.3

C. Other Attributes, Reversed Roles, and Other Pairs

The channel is not one-way: with the roles swapped (7B sender, 3.7B receiver) the ridge map gives 29.0% and the trained projector 63.3%, with the deranged cache at 5.7% and a 63% follow rate—the same pattern at slightly lower accuracy. With Qwen3-4B as the sender (36 layers, 8 KV heads, head dimension 128—matched KV geometry—but a different vocabulary and pre-training), receiver positions are aligned to sender tokens by character offsets and the same recipe is run: top-3 ridge 18.0% (raw injection 15.7%, deranged 13.0%), trained projector 76.7 / 70.3 / 68.0% over three training seeds (mean 71.7, sd 4.5; deranged $\leq 4.7\%$, follow rate 80% for seed 0), SQuAD F1 48.0 / 51.9 / 44.9 (mean 48.3) against 54.7 for the receiver reading the text. The linear map alone does nothing across families. The recipe fails for K2-Horizon-0.9B, a same-family sender with a different vocabulary and head dimension 64: the trained projector reaches 17.0% (deranged 14.0%, follow 11.3%; content transfer 3 points) and 24.2 F1 on SQuAD. K2-0.9B differs from the receiver in vocabulary, layer count and head dimension at once; which difference matters is untested, and cross-head maps exist in other work, so the failure is of this parameterisation, not of the pair.

Table 9. Binding transfer with city and animal attributes ($n = 500$, 8 names) and with the roles reversed (7B sender, 3.7B receiver; 6 names, $n = 300$).

setting	projector	none	project	derange	follow
city	ridge top-3	14.2	19.8	11.8	18.4
city	trained	14.2	61.8	6.4	60.2
city	deranged-sender ctrl	14.2	15.4	11.4	18.6
animal	ridge top-3	12.2	12.8	12.0	11.2
animal	trained	12.2	67.2	5.2	64.6
animal	deranged-sender ctrl	12.2	12.4	12.8	9.8
reversed	ridge top-3	11.7	29.0	12.3	31.3
reversed	trained	11.7	63.3	5.7	63.0

Table 10. Four sender–receiver pairs, same recipe. Noma 6-name variant, $n = 300$, chance 12.5%; percentages. “proj.” is the *project* arm, “der.” the *derange* arm, “follow” the follow rate. SQuAD F1 is for the trained projector ($n = 300$; text ceiling 54.7). K2-Horizon unless named. Qwen3-4B row: proj. and F1 are mean $\pm$ sd over three training seeds (76.7 / 70.3 / 68.0; F1 48.0 / 51.9 / 44.9); der. is the worst seed, follow is seed 0. The raw arm is undefined for K2-0.9B (head dimension 64).

sender $\rightarrow$ receiver	raw	ridge top-3			+ trained residual			F1
		proj.	der.	follow	proj.	der.	follow	
3.7B $\rightarrow$ 7B	14.0	22.0	10.7	30.3	71.3	4.3	71.7	53.0
7B $\rightarrow$ 3.7B	11.7	29.0	12.3	31.3	63.3	5.7	63.0	—
Qwen3-4B $\rightarrow$ 7B	15.7	18.0	13.0	17.0	71.7 $\pm$ 4.5	≤ 4.7	80.3	48.3 $\pm$ 3.5
0.9B $\rightarrow$ 7B	n/a	13.3	13.0	10.7	17.0	14.0	11.3	24.2

D. Different Contexts: HotpotQA

Table 11. HotpotQA bridge questions, $n = 400$. Sender and receiver each hold one of the two supporting paragraphs; the receiver always reads its own paragraph and the question. Token-F1; the mapped cache is mean $\pm$ sd over three training seeds (37.9 / 39.7 / 42.1), the deranged cache their range.

receiver additionally gets	F1
nothing (own paragraph + question)	30.1
the sender’s paragraph as text	45.8
the sender’s paragraph as a mapped cache	39.9 $\pm$ 2.1
a deranged mapped cache	29.9–32.1
the sender’s paragraph through the ridge map	26.3
a zero cache	20.9

This is the LCF-X setting of Rossi et al. (2026). The receiver with its own paragraph and the question reaches F1 30.1; with both paragraphs as text 45.8; with its own paragraph plus the sender’s mapped cache 37.9 / 39.7 / 42.1 over the three training seeds (mean 39.9, sd 2.1)—63% of the gap—while the deranged cache gives 29.9–32.1 (no help), the ridge map 26.3 and a zero cache 20.9. The channel carries the *other* paragraph, and only the trained map does.

E. Distance Sweep and Layer Bands

F. Compression Sweep

A projector with bottleneck 32 has 70 M parameters and keeps SQuAD F1 (53.7) while losing 9 points on the binding test (64.3%); bottleneck 16 (62 M) gives 61.3% and 51.6 F1. The filler rows are the structural null of Section 7: the receiver’s own cache of a neutral text is a valid, episode-free state, so a collapse under it is missing content rather than broken attention.

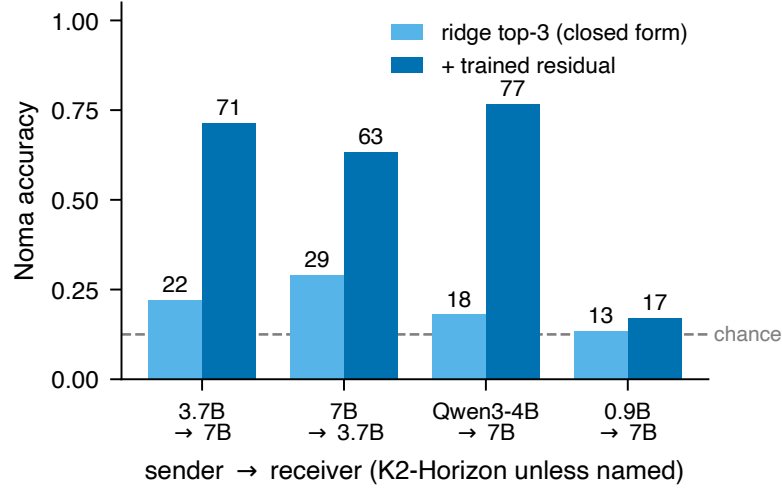


Figure 4. Noma accuracy (*project* arm) for the closed-form ridge map and the trained projector on four sender–receiver pairs ($n = 300$, 6-name variant). Dashed line: chance. The trained map crosses families (Qwen3-4B) but not a vocabulary and head-dimension change within the family (K2-0.9B).

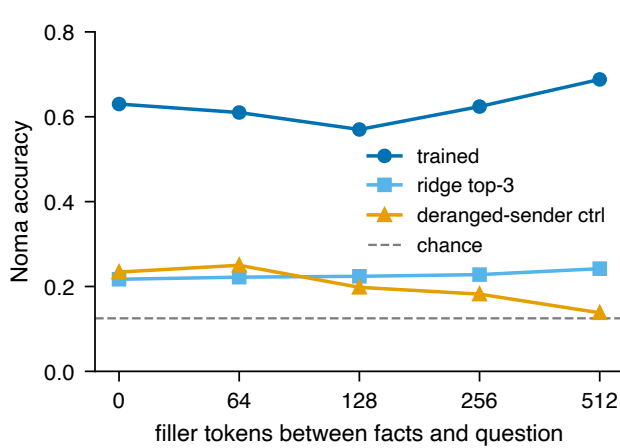
Table 12. Noma accuracy (*project* arm) versus filler tokens between facts and question. 8-name variant; $n = 1000$ at pad 0, $n = 500$ otherwise.

projector	0	64	128	256	512
ridge top-3	21.7	22.2	22.4	22.8	24.2
trained	63.0	61.0	57.0	62.4	68.8
deranged-sender ctrl	23.4	25.0	19.8	18.2	13.8

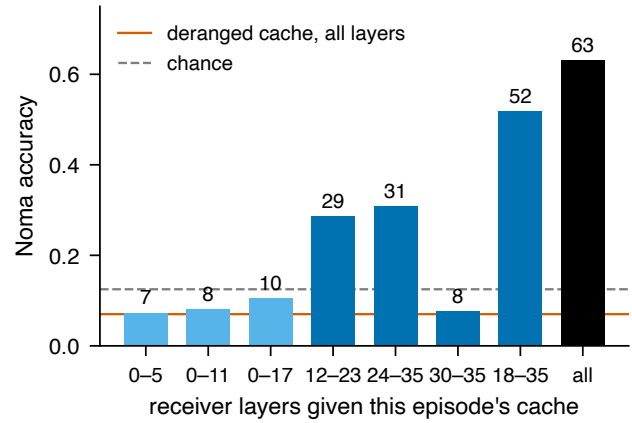
G. When Should the Small Model Hand Off?

On 600 SQuAD items the 3.7B alone reaches F1 52.2 and gets 35.5% of items wrong ($F1 < 0.5$). A logistic probe on its pre-action hidden state (last prompt token, layers 12/18/24; the probe of Kossen et al., 2024 with a failure label; five-fold, not grouped by passage, $C = 0.1$) predicts those failures with AUROC .749, better than its own semantic entropy (.689), sample disagreement (.683), max-probability (.582) or entropy (.584) (Table 14). But the 7B reading the passage reaches only 55.7, the mapped cache 51.8 and the verbal handoff 47.0, so no trigger makes a handoff pay at equal latency (Figure 7): the probe-triggered text handoff climbs from 52.2 to 55.7 as the handoff rate goes to 1, the cache handoff stays flat at 52, and only an oracle trigger gains (56.4 at rate .4). On this task the receiver is not enough better than the sender for a handoff of any kind to be worth it; the probe result stands, the policy result waits for a task with a real gap.

H. Additional Figures



(a) Accuracy against filler tokens between the facts and the question (8-name variant; $n = 1000$ at 0, $n = 500$ otherwise). No decay; the best value is at the training prefix length of 512.



(b) Accuracy when only the labelled band of receiver layers receives this episode's mapped cache (the deranged episode's cache fills the rest; Table 2). Solid line: the deranged cache in all layers; dashed: chance.

Figure 5. Distance sweep and layer-band injection.

Table 13. Every compression of the seed-2 projector's mapped cache. KB per token is for 512-token SQuAD passages; low-rank truncation and mean-vector fill have a per-sequence overhead, so their per-token size depends on sequence length (for the 29-token Noma facts, rank 32 exceeds the full cache). Noma: 6-name variant, $n = 300$, percent; SQuAD: $n = 300$, token-F1. "filler" rows fill the untransmitted layers or heads with the receiver's own cache of a neutral text of the same length.

spec	KB / token	Noma project	Noma derange	SQuAD F1
none (full, bf16)	144	75.3	2.3	53.0
int8	72	76.0	2.7	52.1
int4	36	75.0	3.7	51.6
rank 32	78.7	75.0	2.3	45.3
rank 16	39.3	75.0	2.3	29.6
rank 8	19.7	51.7	7.0	17.2
layers 18–35, zeros	72	24.3	9.7	8.3
layers 18–35, zeros, int8	36	24.7	9.7	7.8
4 heads, zeros	72	25.7	8.7	8.2
2 heads, zeros	36	22.3	9.7	5.9
layers 12–35, mean fill	96.4	35.0	11.0	28.8
layers 18–35, mean fill	72.7	33.7	11.7	21.4
layers 18–35, mean fill, int4	18.7	33.3	11.7	21.1
layers 24–35, mean fill	48.9	29.3	12.0	14.5
4 heads, mean fill	72.7	23.0	11.3	15.3
layers 12–35, filler	96	30.7	11.7	18.9
layers 18–35, filler	72	28.7	12.0	16.1
layers 18–35, filler, int4	18	30.0	11.7	16.9
layers 24–35, filler	48	25.3	10.0	13.3
4 heads, filler	72	26.7	10.0	12.8
projector bottleneck 32 (70 M params)	144	64.3	5.7	53.7
projector bottleneck 16 (62 M params)	144	61.3	7.7	51.6

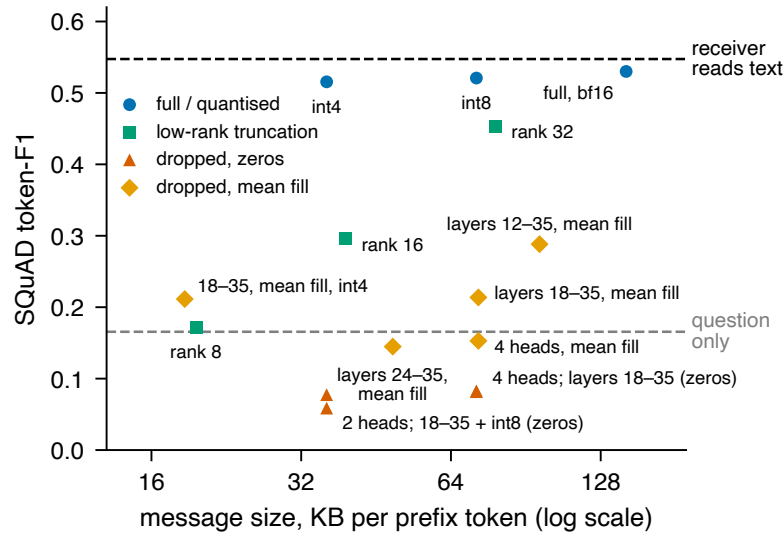


Figure 6. Message size against SQuAD F1 for every compression of the mapped cache in the zero- and mean-fill sweep ($n = 300$). Quantisation is free; low-rank truncation and dropping layers or heads are not. The two “zeros” points at 72 and 36 KB each hide two coincident specs.

Table 14. Handoff policy on 600 SQuAD items. Top: AUROC of each trigger at predicting the sender’s own failures ($F1 < 0.5$). Bottom: F1 of each arm when every item is handed off (rate 1).

sender (3.7B) alone: F1 52.2, wrong on 35.5% of items	
trigger	AUROC
probe on the sender’s hidden state	.749
semantic entropy (10 samples)	.689
sample disagreement	.683
max-probability	.582
entropy	.584
receiver (7B) sees	
F1 at rate 1	
the passage as text	55.7
the mapped cache	51.8
the sender’s answer + confidence word	47.0

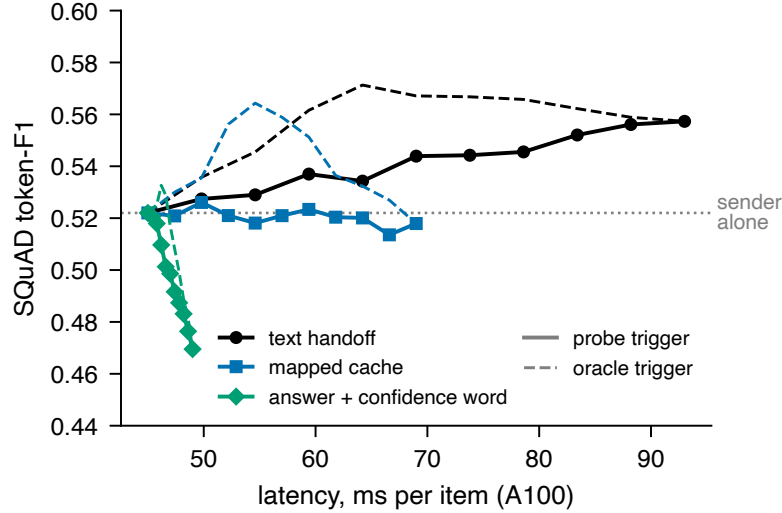


Figure 7. F1 against latency per item as the handoff rate goes from 0 (sender alone, 45 ms) to 1, for the three handoff arms under the probe trigger (solid) and an oracle trigger that hands off exactly the sender’s failures (dashed). 600 SQuAD items. Only the oracle gains; the probe-triggered text handoff climbs to the receiver’s own 55.7, the cache handoff stays flat.

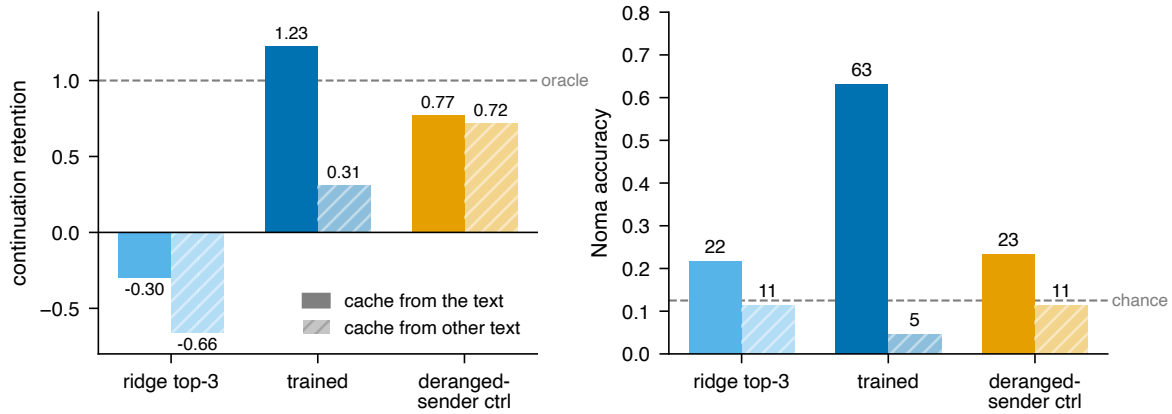


Figure 8. Retention and the binding test disagree about the deranged-sender control. Left: continuation retention on 128 fresh held-out sequences for the three projectors, with the sender reading the true prefix (solid) or a swapped one (hatched). Right: Noma accuracy for the same three projectors ($n = 1000$, 8-name variant), *project* arm (solid) and *derange* arm (hatched). Retention ranks the deranged-sender projector at 0.77, most of which survives a swapped sender; the binding test puts it at chance.

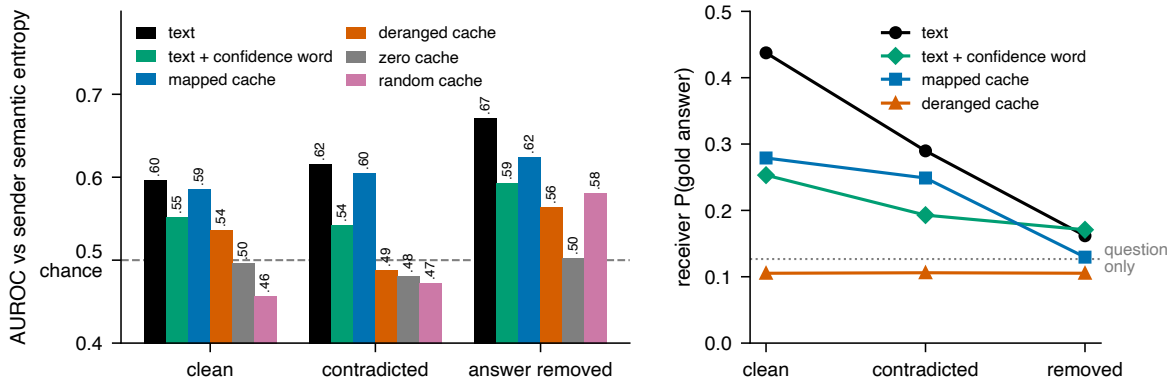


Figure 9. Left: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, per handoff and passage variant (377 items), including the zero and moment-matched random caches. Right: the receiver’s $P(\text{gold})$ from clean to contradicted to removed passages. The cache path is less confident than the text path on clean passages and keeps the original answer ahead under a contradiction.